

MooreVIEW Opta Firmware & Operation

Firmware version: 2.3.45

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Operator and integrator reference — MQTT Parc ST runtime on Arduino Opta

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MooreVIEW Opta — Firmware & Operation Reference

Sketch: `MooreviewOptaMqttSt`
Current version: `2.3.45` (`mv_version.h`)
Protocol: MQTT Parc `mooreview/v1/{deviceId}/...`

MooreVIEW PC integrates via driver type `mqtt_parc` . In-app help: **F1** → **MQTT Parc hub & Opta**.

Overview

MooreviewOptaMqttSt combines an on-device **ST bytecode runtime** with **MQTT Parc** telemetry and remote deploy. The PC hub ingests telemetry into `data/parc.json` ; MooreVIEW deploys programs with `put_program` , runs ST with `runtime_start` , and syncs forces and time over MQTT.

Important: Parc deploy updates the ST program and NV storage only. It does **not** flash this sketch. After firmware changes in git, re-upload via Arduino IDE.

Hardware

Component	Details
Base	Arduino Opta — 8 digital inputs (I1–I8), 4 relays (R1–R4), raw analog I1_RAW–I8_RAW
Relay LEDs	On-board LEDs D0–D3 mirror relay states R1–R4
Expansion slot 1	Closest to base; up to 5 slots configured
AFX00005 (D1608E)	16 digital inputs + 8 relay outputs
AFX00007 (A0602)	8 analog channels + 4 PWM outputs
Identity	ATECC608B serial → deviceId = opta_ + 18 hex digits
Network	Ethernet (primary); optional WiFi setup AP

Requires **24 V** on expansion modules for detection. Install **Arduino_Opta_Blueprint** for expansion scan.

Firmware architecture

Module	Role
MooreviewOptaMqttSt.ino	Main loop, HTTP routes, deferred NV auto-run
mv_mqtt.cpp	Parc MQTT client, command dispatch, telemetry
mv_st.cpp, mv_bc.cpp	ST bytecode VM (same AST as MooreVIEW PC)
mv_tags.cpp, mv_io.cpp, mv_expansions.cpp	Tag model, physical I/O, expansion tags
mv_program_store.cpp	QSPI NV program file, CRC, auto-run flag
mv_store.cpp, mv_setup_web.cpp	Device config KV (Ethernet, broker, expansions)
mv_io_map.cpp	/io-map web UI and JSON API
mv_rtc.cpp	Software wall clock + queued HAL RTC
mv_identity.cpp	ATECC608 → deviceId
mv_http.cpp	Native Ethernet HTTP server
mv_ota.cpp, mv_version.cpp	OTA upload, version reporting

Serial debug: USB **115200**. Milestone logs use `[MV*]` prefix (always on).

Network

Ethernet

DHCP or static IP configured on `/setup`. Default static example: `192.168.1.50`.

MQTT broker

- **Topic prefix:** `mooreview/v1`
- **Telemetry:** `mooreview/v1/{deviceId}/telemetry`
- **Commands:** `mooreview/v1/{deviceId}/cmd`
- **Responses:** `mooreview/v1/{deviceId}/cmd/response`
- **Config:** `mooreview/v1/{deviceId}/config` (pause telemetry during debug)

Broker IP/port saved in device NV on `/setup` (example: `192.168.1.233:1883`). Must match MooreVIEW **System setup** → **MQTT broker URL** (PC LAN IP, not `127.0.0.1` from the device). Reboot after broker change.

PC hub

Enable **MQTT Parc hub** and set broker URL in MooreVIEW System setup. Start Mosquitto on the PC LAN interface (`npm run mqtt:start`).

Web pages and APIs on device

Method	Path	Purpose
GET	<code>/</code> , <code>/setup</code>	Setup GUI — Ethernet, WiFi AP, MQTT broker, expansions, ST status
GET	<code>/io-map</code>	Live I/O map (base + expansions); Enable I/O update polls API
GET	<code>/api/status</code>	Full health JSON
GET	<code>/api/status/lite</code>	Lightweight status (setup panel refresh)
GET	<code>/api/tags</code>	Tag snapshot
GET	<code>/api/io-map</code>	I/O map JSON (logic vs effective when forced)
GET/PUT	<code>/api/setup/config</code>	Read/write persisted device config
POST	<code>/api/setup/scan</code>	Scan expansion modules
POST	<code>/api/setup/reboot</code>	Reboot device
PUT/DELETE	<code>/api/program</code>	Local AST deploy / clear (engineering)
GET/PUT	<code>/api/program/autorun</code>	NV auto-run flag
POST	<code>/api/runtime/start</code> , <code>/api/runtime/stop</code>	Local runtime control
GET	<code>/api/ota</code>	OTA status
POST	<code>/api/firmware</code>	Flash .bin (reboots)

Navigation bar links **Setup** ↔ **I/O Map** on both pages.

MQTT command reference

Publish JSON to `mooreview/v1/{deviceId}/cmd` :

```
{ "id": "<uuid>", "op": "<command>", "body": { ... } }
```

Command	Body (key fields)	Effect
put_program	bc, tagIds, tags, protocolVersion, clientVersion, optional autoRunOnBoot	Load bytecode, save to QSPI NV, register tags
runtime_start	scanMs (default 100)	Start ST scan loop
runtime_stop	—	Stop scan loop
runtime_status	—	Return running state, programOk
set_force	tagId, forceInput, forceOutput, forceValue	Apply PLC-style force
clear_force	tagId	Release force on tag
sync_time	unixUtc, tzOffsetMin	Set software wall clock (not during deploy)
set_autorun	enabled or autoRunOnBoot	Persist auto-run on power-up (KV)
clear_program	—	Clear program, stop runtime
get_program	—	NV status: programFromNv, programNvCrc, autoRunOnBoot
scan_expansions	—	Rescan modules, refresh expansion tags

`put_program` blocks other commands until complete. Minimum firmware for Parc deploy: **v2.3.18+**. Recommended: **v2.3.41+** (reliable MQTT cmd subscribe).

NV storage

Store	Location	Contents
Device config KV	/kv/mv_device_cfg	Ethernet, WiFi AP, expansion slot types, MQTT broker IP/port
Auto-run flag KV	/kv/mv_autorun	autoRunOnBoot (0/1)
Program file	QSPI FAT /fs/mv_program.bin	Header (name, CRC, auto-run) + bytecode

Program NV uses Modbus CRC16 over bytecode (matches PC `bcDeployCrc`). On boot, firmware loads NV program before MQTT; if `autoRunOnBoot` is set, ST starts ~500 ms after MQTT connects.

ST program lifecycle

```
Power-up → load NV program (if present) → Ethernet + MQTT connect
        → [autoRunOnBoot?] defer runtime_start ~500 ms
PC Connect → runtime_status + sync_time
PC Download & Start → put_program (skip if NV CRC matches) →
runtime_start
Deploy → save bytecode to QSPI (async write phases)
Stop → runtime_stop (program remains in NV)
clear_program / DELETE /api/program → clear NV program
```

MooreVIEW **System setup** → **Auto-run ST on Opta after power-up** sends `autoRunOnBoot: true` with deploy. PC boot auto-start waits ~8 s before deploy when Remote is on.

Force and I/O

- **Mux model:** `logicValue` = hardware read; effective tag getters return forced value when `forceInput` or `forceOutput` is set.

- **PC Tags** → **Force** sends `set_force` / `clear_force` over MQTT when Remote execution is on.
- **Relay outputs:** R1–R4 drive relays and mirror to LED_D0–LED_D3.
- **I/O map:** `/io-map` shows forced points highlighted; enable polling with **Enable I/O update**.
- **Expansions:** Tags auto-created after scan (e.g. expansion DI/DO/AI naming from slot index).

Time sync

- PC sends `sync_time` on **Connect** and **daily** for linked Opta drivers.
- Firmware sets an immediate **software wall clock** for timestamps and queues HAL RTC when idle (HAL write can block ~15 s — not done during deploy).
- `put_program` does **not** sync time.
- Before sync, serial logs use uptime; after sync, `YYYY-MM-DD HH:MM:SS` . Setup **Clock** row shows `rtcTime` .

Version history (2.3.30+ highlights)

Version	Notes
2.3.45	Current release (<code>mv_version.h</code>)
2.3.41+	MQTT cmd subscribe reliability; fix telemetry-OK-but-cmd-timeout
2.3.30–2.3.40	NV program store, skip-deploy CRC, I/O map web UI, expansion telemetry
2.3.24+	Native HTTP, RTC software clock, broker NV on <code>/setup</code>
2.3.18+	Minimum for MQTT <code>put_program</code> deploy from MooreVIEW
2.3.11+	Fix 2.3.8 stack overflow in MQTT cmd handler
2.3.8	Do not use — ~20 KB stack buffer broke all MQTT

Always verify with GET `/api/status` → `firmwareVersion` .

PC integration checklist

- ☐ Flash **MooreviewOptaMqttSt v2.3.41+** via Arduino IDE
- ☐ Set MQTT broker on Opta `/setup` (LAN IP, port 1883) → Save → Reboot
- ☐ Start Mosquitto on PC LAN (`npm run mqtt:start`)
- ☐ **System setup:** enable MQTT Parc hub, broker URL, Remote ST execution
- ☐ Add `mqtt_parc` driver (template or bulk-add from Parc registry); `deviceId` matches firmware
- ☐ **Sync tags from device** on driver card; **Scan expansions** if modules fitted
- ☐ Optional: **Auto-run ST on Opta after power-up** in System setup
- ☐ **Program** → **Remote** on → **Download & Start**
- ☐ Optional: assign **position ID** for plant location; use **Replace hardware** for change-outs
- ☐ Commissioning: **Tags** → **Force** (syncs to Opta); **Live I/O** → **Enable I/O update**
- ☐ Operations: **Report** → **MongoDB logs** for system log and hardware history

Troubleshooting

Symptom	Likely cause	Action
No telemetry	Broker/firewall/Mosquitto	Match PC + Opta broker IP; check LAN routing
Telemetry OK, cmd timeout	Old firmware or subscribe fail	Reflash v2.3.41+ ; Serial: MQTT subscribed cmd+config
Broker mismatch	<code>127.0.0.1</code> on device	Set PC LAN IP on <code>/setup</code> and System setup
Deploy fails / old fw	Firmware < 2.3.18	Arduino IDE upload MooreviewOptaMqttSt
HTTP dead during deploy	Large put on old fw	Use MQTT deploy; upgrade firmware
ST not running after link	Linked ≠ running	Download & Start , not Connect alone
Wrong deviceId	Manual id typo	Use Add from Parc registry or read <code>/setup</code>
NV program not auto-run	Flag off or no deploy	Deploy once; enable auto-run on PC and/or device
Clock shows uptime	No sync yet	Connect driver or wait for daily sync_time
Expansions missing	24 V / library	Seat modules in slot 1; install Opta Blueprint; Scan

Serial always-on: `[MV*]` boot, subscribe, cmd lines @ **115200**.

Related docs

- Flash/build: [README.md](#)
- PC MQTT Parc: `docs/MQTT_PARC.md` , `docs/BASELINE_TEST.md`
- ST programs: `st/opta-mqtt/README.md`